

Easy and Secure File Transferring with WinSCP

Working in a multi-OS environment often requires transferring files between two machines running entirely different operating systems. Even home users who like a range of OSs running on their machine, face the same situation. Although there are numerous techniques available, from using a pen drive to Samba, if you're looking for easy-to-use solutions, consider WinSCP and VSFTPD.

WinSCP is a decent tool for Windows users, allowing them to share files with their Linux counterparts with ease. So before demonstrating how to use it, first install it on your system.

Obviously, the two systems must be on the same network, with proper IP addresses, etc. In my case, I'm running RHEL 6 on top of Windows 7 using VMware. So here are the steps to connect the Linux in the VM to Windows on the host. If your Windows and Linux systems are different physical machines, VMware users can skip this part.

- Open your virtual machine and change the Ethernet setting to *Custom* and select *VMnet8(NAT)*.
- Assign an IP address to your machine (mine is the C-class 192.168.0.1).
- Now switch to Windows and open *Network and Sharing Center* and select *Change adapter settings*, and here you'll see a number of adaptors listed. Select the adaptor for VMnet8 (on my system, it's named *Local Area Connection 3*). Right-click it and select *Properties*, and again select *Internet Protocol Version 4 (TCP/IPv4)*. Then click on *Properties* (see Figure 1).
- Assign an IP address of the same class as that of our Linux machine (mine is 192.168.0.2).
- Now the configuration is complete, and you can verify it with *ping*. You can check the IPs with the *ifconfig* (Linux) and *ipconfig* (Windows) commands.

Now it's time to start WinSCP and see it in action. On the starting screen, enter *Host name*, *User name* and *Password* and click *Login* (Figure 2). On connecting to a machine for the first time, it'll generate a warning about the authenticity of the machine you are trying to connect to. Just click *Yes*, and it'll connect to the machine. Also copy the key to its cache, so that the next time you log in to the same machine, you don't see the same warning.

After a successful login, you'll see a screen divided in two panels, as in Figure 3. The left panel shows the files on the Windows machine, and the right one shows the files on your



Linux machine. Now you can copy/move files with simple drag-drops, or using the keyboard shortcuts mentioned at the bottom of the screen. To select multiple files to copy/move, hold the control key while selecting the files/directories.

Securing your FTP server

FTP is widely used for file transfers, and here we'll deploy an FTP server with OpenSSL support to make file transfers more secure. The steps are:

- Install the *vsftpd* package by issuing the `yum install vsftpd` command.
- Install *openssl* by issuing the `yum install openssl*` command.
- Change directories: `cd /etc/vsftpd` and create a self-signed certificate (run `openssl req -x509 -nodes -days 365 -newkey rsa:1024 -keyout vinayak.pem -out vinayak.pem`). Here, you can use any name for your certificate file, like I've used *vinayak.pem*. Once you issue this command, you'll be asked to answer a few queries, after which the certificate file will be created.
- Open `/etc/vsftpd/vsftpd.conf` file in a text editor and add the following lines at the end of the file:

```
ssl_enable=YES
allow_anon_ssl=YES
force_local_data_ssl=YES
force_local_logins_ssl=YES
ssl_tlsv1=YES
```